Outcomes After Comprehensive Vision Rehabilitation Using Vision-related Quality of Life Questionnaires: Impact of Vision Impairment and National Eye Institute Visual Functioning Questionnaire

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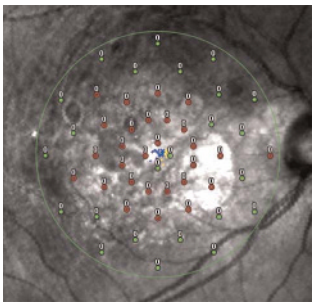
SIGNIFICANCE: This research is significant because, although vision-related quality of life (VRQoL) is improved after vision rehabilitation (VR), patients with certain characteristics respond less positively on VRQoL measures, and this should inform future care.

PURPOSE: The purposes of this study were to evaluate how two VRQoL questionnaires compare in measuring change in patient-reported outcomes after VR and to determine if patient characteristics or occupational therapy (OT) predict higher scores after rehabilitation.

METHODS: In a prospective clinical cohort study, 109 patients with low vision completed the Impact of Vision Impairment (IVI) and the National Eye Institute Visual Functioning Questionnaire (NEI VFQ-25) before and after VR. Comprehensive VR included consultation with an ophthalmologist and OT if required. The relationships of six baseline characteristics (age, sex, visual acuity, contrast sensitivity, field loss, diagnosis) and OT were assessed with VRQoL scores using multivariable logistic regression.

RESULTS: The mean (SD) age was 68.5 (19.2) years, and 61 (56%) were female. After rehabilitation, increases in scores were observed in all IVI subscales (reading [$P < .001$], mobility [$P = .002$], well-being [$P = .0003$]) and all NEI VFQ-25 subscales (functional [$P = .01$], socioemotional [$P = .003$]). Those who were referred to OT but did not attend and those who had hemianopia/field loss were less likely to have higher VRQoL in IVI mobility and well-being. Those attending OT for more than 3 hours were less likely to have better scores in emotional NEI VFQ. Men were less likely to have increased scores in functional and emotional NEI VFQ, whereas those with diagnoses of nonmacular diseases had higher odds of having increased scores on the emotional NEI VFQ (all, $P < .05$).

CONCLUSION: Both the IVI and the NEI VFQ-25 detected change in patients' VRQoL after rehabilitation. Most of the patient characteristics we considered predicted a lower likelihood of increased scores in VRQoL.



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Optom Vis Sci 2019;96:87–94. doi:10.1097/OPX.0000000000001327
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Vision loss is a global health problem affecting 36 million who are blind and 217 million who are visually impaired.[1] Vision rehabilitation is an option for those whose vision cannot be improved with treatment, surgery, or refractive correction, also referred to as low vision. Close to 4 million Americans have low vision or blindness.[2] Vision rehabilitation aims to optimize patients' functional ability and quality of life with assistive devices, training, adaptive strategies, and psychosocial supports. Interventions vary depending on degree and pattern of vision loss.[3]

There is great interest in measuring outcomes of vision rehabilitation to assess effectiveness. Patient-reported outcomes are commonly used[4,5]; however, there is no consensus about best questionnaires.[6] It has also become pertinent to understand if patient's sociodemographic or clinical variables predict who benefits most and least from rehabilitation. To address these questions, this study compared results of two commonly administered, validated vision-related quality-of-life questionnaires before and after rehabilitation, and evaluated the relationship of baseline characteristics as well as completion of occupational therapy to change in patient-reported outcomes in a clinical population of low-vision patients undergoing vision rehabilitation.

The two patient-reported outcome measures used in our study were the National Eye Institute Visual Functioning Questionnaire,

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one of the most commonly used vision-related quality of life questionnaires in ophthalmic clinical research, and the Impact of Vision Impairment, a questionnaire specifically developed to measure outcomes from vision rehabilitation. Both questionnaires are used in vision rehabilitation research[4,5]; however, to our knowledge, they have never been evaluated together longitudinally in a low-vision population. Knowing which questionnaire is most sensitive to detect changes after rehabilitation may assist with future questionnaire selection for vision rehabilitation research. Based on vision rehabilitation literature and previous psychometric studies,[7,8] we hypothesize that the Impact of Vision Impairment and National Eye Institute Visual Functioning Questionnaire will differ in vision-related quality-of-life measurement after vision rehabilitation in a population of low-vision patients.

As part of usual care, the Vision Rehabilitation Centre at Massachusetts Eye and Ear collects many patient variables: demographics, visual function, diagnosis, field loss testing using a macular perimeter, and occupation therapy progress notes. Based on our clinical experience, we hypothesize that older patients with worse fields, worse visual function, and less occupational therapy would demonstrate less improvement in vision-related quality of life compared with their counterparts. Understanding which patient characteristics are associated with improved patient-reported outcomes can inform effective models of care.

METHODS

Recruitment

This prospective clinical cohort study was conducted at the Vision Rehabilitation Clinic of Massachusetts Eye and Ear Infirmary from December 2012 to December 2014. Patients attending the clinic for the first time were invited to participate. Participants were older than 18 years, English speaking, and without severe hearing or cognitive impairment as assessed by a clinician. Written consent was obtained from all participants. The research study was approved by the institutional review board at the Massachusetts Eye and Ear Infirmary and complied fully with the tenets of Declaration of Helsinki.

Comprehensive Vision Rehabilitation

The Vision Rehabilitation Clinic at the Massachusetts Eye and Ear Infirmary provides a multidisciplinary model of comprehensive vision rehabilitation that addresses five areas of patient function: reading, activities of daily living, safety, continuing to participate in activities despite vision loss, and psychosocial well-being (American Academy of Ophthalmology, 2013).[3] All subjects had an assessment by an ophthalmologist (MLJ) who is a specialist in vision rehabilitation and were referred to occupational therapy if goals identified on the initial consultation could be addressed with training and/or training with devices. Occupational therapy included up to 11 hours of therapy at home or in the clinic and focused on addressing the patients' individual goals with appropriate training and devices, such as training patients to navigate pages of text with a magnifying device, using alternate strategies such as operating an audio book player, or using a computer with accessibility options. Occupational therapists also provided referral to other multidisciplinary services such as orientation and mobility training to use a long white cane. As appropriate, patients were offered resources and referral to support groups, counselors, and social work services.

Baseline Characteristics

Six baseline sociodemographic and clinical characteristics were extracted from patients' medical records: (1) age, (2) sex, (3) visual acuity, (4) contrast sensitivity, (5) visual field loss, and (6) diagnosis of eye condition. An additional variable, whether the subject completed occupational therapy, was also included. Best-corrected visual acuity in the better eye was measured using the Early Treatment Diabetic Retinopathy Study chart at 1 m or a projected chart (in logMAR). Binocular contrast sensitivity was measured using the Pelli-Robson chart at 1 m. Extent of central field loss was measured monocularly using the OPTOS scanning laser ophthalmoscope macular perimeter OPKO/OTI (OPKO Instrumentation, Miami, FL). The extent of the scotoma was estimated using monocular perimetry tests and by determining the scotoma that would overlap, viewing binocularly.[9] Peripheral field loss was obtained from patient notes.

Administration of Questionnaires

All patients were mailed the Impact of Vision Impairment questionnaire and National Eye Institute Visual Functioning Questionnaire 2 weeks before their baseline appointment at the clinic and asked to complete the questionnaires themselves or with the assistance of others (family, friends, or clinic staff). Post-rehabilitation questionnaires were sent to subjects 1 month after discharge or within 3 months after last occupational therapy appointment for those who did not return for further occupational therapy treatment. Subjects were followed up by telephone if questionnaires were not returned, and questionnaires were administered by telephone if subjects wished it so. Administration of questionnaires was performed by the clinic research coordinator.

National Eye Institute Visual Functioning Questionnaire

The National Eye Institute Visual Functioning Questionnaire is a vision-related quality-of-life questionnaire that has been previously validated in patients with eye diseases (including age-related macular degeneration and low vision).[7,8,10] The original questionnaire scoring has 12 vision-specific subscales.[7–9] The driving subscale was not used in the current study, and three additional questions about near vision (A3, A4, and A5) were included in the hope that additional questions will help measure improvement in patients with central vision loss after rehabilitation.

Impact of Vision Impairment Questionnaire

The Impact of Vision Impairment is a vision-related quality-of-life questionnaire that has been previously validated in a low-vision population.[11] The questionnaire has a total scale[12,13] and three subscales: reading and accessing information (reading), mobility and independence (mobility), and emotional well-being (well-being).

Psychometric Assessment of the Outcome Measures

Rasch analysis was used to assess the psychometric properties of the Impact of Vision Impairment and the National Eye Institute Visual Functioning Questionnaire using the Andrich rating scale model[14] with Winsteps software (version 3.81; Chicago, IL).[15] Rasch analysis is a form of item response theory and enables ordinal raw questionnaire scores to be transformed to estimates of interval measures (expressed in log of the odds units, or logits), thus enabling subsequent parametric analyses to be confidently performed.[16] In the case of the Impact of Vision Impairment and the National Eye Institute Visual Functioning Questionnaire, a high

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person measure (in logits) indicates that a person possesses a high level of the assessed latent trait (e.g., vision-related quality of life). Rasch analysis also provides significant insight into the psychometric properties of the scale, including (*a*) appropriate use of response categories, (*b*) measurement precision, (*c*) how well items "fit" the underlying trait, (*d*) unidimensionality, (*e*) targeting of item difficulty to patients' ability, and (*f*) differential item functioning.[17]

To generate valid pre-rehabilitation and post-rehabilitation person measures, the data were anchored to item measures and structure calibrations at baseline for the subsequent analyses, which is standard in pre/post–Rasch analyses.[18,19] This approach reduces distortion of the estimated person measures by anchoring the Impact of Vision Impairment questionnaire and National Eye Institute Visual Functioning Questionnaire to baseline, thus ascertaining that the same instrument with identical measurement properties is used to repeatedly test participants over time. Baseline and follow-up data were also stacked to establish that differences between the scores for all outcome measures at baseline and follow-up were valid indicators of changes over time.[20]

Overall, the composite Impact of Vision Impairment scale displayed suboptimal fit to Rasch model parameters. Although scale precision and targeting were excellent, there was evidence of multidimensionality, which could be resolved only by splitting the Impact of Vision Impairment into three separate scales: reading (items 1, 3, 5 to 9, 14, and 15), mobility (items 2, 4, 10 to 13, and 16 to 20), and well-being (items 21 to 28). These three scales—which were proposed in the original validation studies of the Impact of Vision Impairment[21,22]—were unidimensional, displayed excellent precision and targeting, were free from differential item functioning, and were used for our subsequent analysis.

Previous Rasch analysis of the National Eye Institute Visual Functioning Questionnaire has demonstrated that the composite scale and 6 of the 12 proposed subscales are dysfunctional, and a two-factor structure—visual functioning and socioemotional—has been proposed.[7,8] Guided both by this prior analysis and by the standardized residual loadings in principal components analysis from our own data, we used a similar two-factor structure. The National Eye Institute Visual Functioning Questionnaire has seven different rating scales for patients to record their responses. Our analysis found that rating scale 6 (items 17 to 19, all of the time to none of the time) and rating scale 7 (items 20 to 25, definitely true to definitely false) were disordered. To resolve disordered thresholds, categories 4 and 5 were collapsed for rating scale 6, and category 3 ("not sure") was coded as missing data in rating scale 7, based on empirical evidence from a similar validation study of the National Eye Institute Visual Functioning Questionnaire using Rasch analysis.[7,8] Rasch analysis of the National Eye Institute Visual Functioning Questionnaire also revealed that the scale was multidimensional, with a high eigenvalue (3.1) and seven misfitting items. The visual functioning scale (items 2, 4, 5 to 14, 19, and A3 to A5) displayed good precision and targeting; however, multidimensionality was evident, with a high eigenvalue (2.7) and two misfitting items. There was also evidence of differential item functioning for sex for item A5 ("how much difficulty do you have doing things like shaving, styling your hair, or putting on makeup"). This item was more difficult for men than for women, despite the two groups having a similar underlying level of visual functioning. Therefore, we iteratively deleted items A3 to A5 to resolve multidimensionality and differential item functioning, and we deleted item 4 because of misfit. The remaining 12 items displayed excellent fit to Rasch model parameters. The socioemotional scale (items 3, 17, 18, and 20 to 25, which relate to the original dependency, mental health, and role limitations subscales) displayed excellent psychometric properties and did not require any amendments to its structure. Thus, the two subscales of visual functioning and socioemotional were used for subsequent analysis.

Statistical Analyses

All statistical analyses were performed using Stata 14.2 for Windows (StataCorp LLC, College Station, TX). Patients' baseline characteristics were reported as mean (standard deviation) for continuous variables and as frequency count (percent) for categorical variables. Independent-sample *t* test and χ^2 test for continuous and categorical variables, respectively, were used to compare baseline characteristics between those who enrolled in the research study and those who declined and between patients who completed both questionnaires and those who did not complete the post-rehabilitation questionnaires. Paired *t* tests were then performed to determine if change in questionnaire scores was significantly different from pre-rehabilitation to post-rehabilitation, with Bonferroni adjustments used for multiple comparisons. A multivariable binary logistic regression model was then used to determine the impact of participants' basic demographic and clinical characteristics, together with occupational therapy attendance and duration, on post-rehabilitation vision-related quality-of-life scores in the subscales of the Impact of Vision Impairment and National Eye Institute Visual Functioning Questionnaire. A two-sided *P* value of less than .05 was considered statistically significant.

RESULTS

Participants

A total of 189 patients agreed to participate in this study and completed pre-rehabilitation questionnaires. Patients (n = 219) who did not agree to participate in this study during the same period did not differ in age, sex, visual acuity, or contrast sensitivity ($P > .05$, data not shown). Only 109 (58%) of the 189 patients completed both pre-questionnaires and post-questionnaires, and the overall mean age (standard deviation) of these participants was 68.5 (19.2) years, with 61 (56%) being female. The reasons for failure to obtain post-rehabilitation questionnaires included (1) subjects not mailing questionnaires after reminders, (2) subjects lost to contact, (3) refusal, and (4) subjects reporting poor health that made it too difficult to complete questionnaires. When baseline characteristics of those who completed both questionnaires and those who did not complete post-rehabilitation questionnaires were compared, the groups did not differ significantly in sex ($P = .42$, χ^2 test), age ($P = .06$), or visual acuity ($P = .08$). However, the group who did not complete post-rehabilitation questionnaires had significantly ($P = .02$) better average contrast sensitivity (mean [standard deviation], 1.30 [0.46]) than did the group who completed both pre-questionnaires and post-questionnaires (mean [standard deviation], 1.14 [0.43]). Table 1 shows the characteristics of patients who completed both pre-rehabilitation questionnaires and post-rehabilitation questionnaires and participated in the study. Patients who were not referred to occupational therapy included those who in the clinician's opinion did not require further training after the initial consultation.

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TABLE 1. Characteristics of patients who completed pre-rehabilitation and post-rehabilitation questionnaires

Variable	n = 109
Age (y), mean (SD)	68.5 (19.2)
Female, n (%)	61 (56.0)
Visual acuity in the better eye, mean logMAR (SD)	0.52 (0.41)
Contrast sensitivity in both eyes, mean log (SD)	1.14 (0.43)
Diagnosis, n (%)	
Macular disease	61 (56.0)
Glaucoma	13 (11.9)
Optic nerve disease	6 (5.50)
RP/peripheral retinal disease	11 (10.1)
Hemianopia	5 (4.59)
Other*	13 (11.9)
Field loss, n (%)	
Central	67 (61.5)
Central and peripheral	29 (26.6)
Peripheral loss	3 (2.75)
Hemianopia	5 (4.59)
No field loss	5 (4.59)
Total hours of OT, n (%)	
<1.5 h	57 (52.3)
≥1.5 h	32 (29.4)
≥3 h	20 (18.3)
OT completion, n (%)	
OT completed	37 (33.9)
OT not completed	47 (43.1)
Referred to OT but did not attend	16 (14.7)
Not referred to OT	9 (8.26)

*Five subjects (4.59%) had other types of field loss from neuro-ophthalmic disorders, two (1.83%) had cerebral palsy, one (0.92%) had keratitis, and another five (4.59%) had complicated diagnoses that involved having several ophthalmic diseases in one or both eyes. OT = occupational therapy; RP = retinitis pigmentosa.

Change in Vision-related Quality-of-life Scores After Rehabilitation

There was a significant increase in vision-related quality-of-life scores on all three Impact of Vision Impairment subscales after rehabilitation: reading (mean change, 0.59 logits; $P < .001$, paired t test), mobility (mean change, 0.38 logits; $P = .002$), and well-being (mean change, 0.68 logits; $P = .0003$; Table 2). Both National Eye Institute Visual Functioning Questionnaire visual functioning and socioemotional subscale scores also increased significantly after rehabilitation ($P = .01$ and $P = .003$, respectively). The effect sizes were moderate for Impact of Vision Impairment reading (Cohen $d = 0.50$) and small for Impact of Vision Impairment well-being (0.37), Impact of Vision Impairment mobility (0.31), and National Eye Institute Visual Functioning Questionnaire visual functioning (0.22) and socioemotional subscales (0.34).[23]

Predicting Change in Vision-related Quality-of-life Scores from Patient Characteristics and Occupational Therapy

Table 3 shows the independent associations of the baseline participant's demographic and clinical characteristics, and attendance and duration of occupational therapy, with pre/post-changes in vision-related quality of life as assessed using the Impact of Vision Impairment and National Eye Institute Visual Functioning Questionnaire. For the demographic variables, only male sex was associated with smaller odds of experiencing an increase in scores in both the functional (odds ratio, 0.29; 95% confidence interval, 0.12 to 0.73; multivariable logistic regression) and emotional (odds ratio, 0.37; 95% confidence interval, 0.15 to 0.92) subscales of the National Eye Institute Visual Functioning Questionnaire. A diagnosis of nonmacular diseases was associated with a higher likelihood of having increased scores in the emotional aspect of the National Eye Institute Visual Functioning Questionnaire (Table 3). In contrast, patients with hemianopia and peripheral field loss had a lower likelihood of having increased scores on Impact of Vision Impairment well-being (odds ratio, 0.008; 95% confidence interval, 0.001 to 0.54). Participants who were referred to but did not attend occupational therapy were less likely to experience a positive change after follow-up in Impact of Vision Impairment mobility scores as compared with those who did not need occupational therapy referrals (odds ratio, 0.04; 95% confidence interval, 0.001 to 0.95).

TABLE 2. Change in vision-related quality-of-life scores of IVI and NEI VFQ-25 questionnaire before and after rehabilitation

Mean (SD) in logits	Pre-rehabilitation	Post-rehabilitation	Pre/post-change	Effect size (Cohen *d*)	*P*
IVI Rasch scores					
Reading	–0.20 (1.22)	0.48 (1.35)	0.59	0.50	<.0001*
Mobility	–0.11 (1.18)	0.26 (1.48)	0.38	0.31	.002*
Well-being	–0.20 (1.85)	0.39 (1.95)	0.68	0.37	.0003*
NEI VFQ-25 Rasch scores					
Visual functioning	0.16 (1.01)	0.38 (1.16)	0.22	0.25	.01
Socioemotional	0.23 (1.39)	0.56 (1.49)	0.34	0.29	.003*

Rasch scores are in logits (range, –4.74 to 4.73). Effect size was calculated by dividing the difference between pre-rehabilitation and post-rehabilitation subscale means over the pooled SD for those means. *$P < .05$ after Bonferroni adjustments. IVI = Impact of Vision Impairment; NEI VFQ-25 = National Eye Institute Visual Functioning Questionnaire.

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TABLE 3. Multivariable* linear regression model of the associations between baseline subject characteristics and change in vision-related quality of life

	IVI			NEI-VFQ	
	Reading, OR (95% CI)	**Mobility, OR (95% CI)**	**Well-being, OR (95% CI)**	**Functional, OR (95% CI)**	**Emotional, OR (95% CI)**
Patient characteristic					
Age	0.98 (0.96–1.01)	0.99 (0.97–1.04)	1.01 (0.98–1.04)	1.00 (0.97–1.02)	1.00 (0.98–1.03)
Male sex	0.87 (0.36–2.13)	0.68 (0.29–1.55)	0.45 (0.18–1.13)	**0.29 (0.12–0.73)**	**0.37 (0.15–0.92)**
Visual acuity	1.32 (0.45–3.89)	1.26 (0.44–3.57)	0.34 (0.09–1.25)	0.68 (0.22–2.08)	0.72 (0.24–2.18)
Contrast sensitivity	0.99 (0.97–1.02)	0.98 (0.95–1.01)	0.97 (0.93–1.01)	0.99 (0.97–1.02)	0.99 (0.97–1.02)
Diagnosis					
Macular disease	Reference	Reference	Reference	Reference	Reference
Others	1.20 (0.28–5.02)	0.84 (0.24–2.91)	1.70 (0.42–6.87)	1.68 (0.43–6.43)	**4.22 (1.02–17.5)**
Type of field loss					
No field loss	Reference	Reference	Reference	Reference	Reference
Central and peripheral	0.49 (0.03–7.30)	0.64 (0.05–7.69)	0.15 (0.004–5.92)	1.05 (0.10–11.3)	0.27 (0.02–4.61)
Central	1.44 (0.09–22.6)	1.00 (0.07–12.9)	0.10 (0.003–3.49)	0.97 (0.10–9.87)	0.39 (0.03–5.92)
Hemianopia and peripheral loss	0.34 (0.02–6.89)	0.82 (0.05–12.2)	**0.008 (0.001–0.54)**	2.22 (0.13–37.8)	0.45 (0.18–11.3)
OT completion					
Not referred	Reference	Reference	Reference	Reference	Reference
Complete, discharged	0.19 (0.02–2.39)	0.09 (0.005–1.93)	7.27 (0.73–72.8)	1.53 (0.21–10.9)	2.37 (0.32–17.5)
Incomplete, no discharge	0.13 (0.01–1.56)	0.07 (0.004–1.54)	3.88 (0.46–32.6)	0.47 (0.07–3.08)	0.85 (0.13–5.56)
Referred, but did not attend	0.13 (0.01–1.72)	**0.04 (0.001–0.95)**	1.04 (0.12–8.99)	0.38 (0.05–2.84)	0.65 (0.09–4.81)
Length of OT					
<1.5 h	Reference	Reference	Reference	Reference	Reference
≥1.5 h	1.52 (0.50–4.61)	1.03 (0.37–2.86)	0.32 (0.09–1.14)	1.28 (0.40–4.06)	0.38 (0.12–1.23)
≥3 h	2.67 (0.68–10.5)	0.98 (0.30–3.29)	0.35 (0.08–1.50)	0.32 (0.08–1.28)	**0.20 (0.05–0.82)**

Bold font indicates significant associations. *Adjusted for all variables in the model. CI = confidence interval; IVI = Impact of Vision Impairment; NEI VFQ-25 = National Eye Institute Visual Functioning Questionnaire; OR = odds ratio; OT = occupational therapy.

Lastly, having attended occupational therapy sessions for at least 3 hours decreased the likelihood of greater vision-related quality-of-life scores in the emotional domain of the National Eye Institute Visual Functioning Questionnaire (odds ratio, 0.20; 95% confidence interval, 0.05 to 0.82). No other associations were found with other participant characteristics and vision-related quality of life.

DISCUSSION

Millions of people in the United States and worldwide have irreversible vision loss, and vision rehabilitation remains the only treatment option for these people. Unlike in most ophthalmology studies, measurement of visual function is not an outcome that reflects the contributions of vision rehabilitation. As a result, patient-reported outcomes in the form of vision-related quality-of-life questionnaires have become the most commonly used outcome measure of rehabilitation.[5] In our study, we measured perceived efficacy of our usual care vision rehabilitation at Massachusetts Eye and Ear Infirmary using two commonly used patient-reported outcomes: Impact of Vision Impairment and National Eye Institute Visual Functioning Questionnaire. We found that indeed our usual care vision rehabilitation led to increased vision-related quality-of-life scores in our patient cohort. Because individual patients respond to vision rehabilitation differently, we examined the data to elucidate whether patient characteristics and participation in occupational therapy predicted a positive change in scores on vision-related quality of life. Surprisingly, sex, field loss, and occupational therapy attendance/duration predicted a less likelihood of higher scores, and only diagnosis other than macular disease was predictive of increases in vision-related quality-of-life scores.

Change in Vision-related Quality-of-life Scores After Rehabilitation

Our main result from the study was that perceived vision-related quality of life increased in all three Impact of Vision Impairment subscales and in the two National Eye Institute Visual Functioning Questionnaire subscales after comprehensive vision rehabilitation. The reading subscale of the Impact of Vision Impairment had the largest effect size (Cohen $d = 0.50$), and the National Eye Institute Visual Functioning Questionnaire socioemotional subscale had a smaller effect size (0.29) than did the similar Impact of Vision Impairment well-being subscale (0.37). Our effect sizes are similar to or a little larger than Impact of Vision Impairment effect sizes reported in published studies, for example, Lamoureux et al.[11] (mobility, 0.17; reading, 0.20; and well-being, 0.30) and

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Wang et al.[24] (mobility, 0.18 and 0.22). Although these previous studies had rehabilitation programs of different intensity from ours, all of the programs aimed to measure outcomes from usual care settings, which ultimately may be similar in their heterogeneity.[11,24] For the National Eye Institute Visual Functioning Questionnaire, only non–Rasch-scored subscale effect sizes ranging from 0.09 to 0.55 (which is similar to our findings) have been published.[25] Although our effect sizes (between 0.2 and 0.5) are on par with most other low-vision rehabilitation outcome studies,[26] several studies have reported larger effect sizes after vision rehabilitation.[27,28] The LOVITT 1 trial by Stelmack et al.[27] had effect sizes between 1.14 and 2.51 using the Veterans Affairs Low-Vision Visual Functioning Questionnaire patient-reported outcomes. The study offered intensive multidisciplinary rehabilitation and provided low-vision devices at no cost to a sample of largely white, male veterans. Because it is unlikely to encounter such a homogeneous sample and uniform treatment paradigm in typical clinical practice, it is reasonable to assume that effect sizes for general vision rehabilitation are much smaller, and in fact, this has been the case in most studies. The effect sizes in the study by Goldstein et al.[28] range between 0.38 and 0.82 using Activity Inventory patient-reported outcomes. Our effect size range of 0.25 to 0.50 has a good concordance with that of Goldstein et al.[28] as well as similar follow-up rates, which may contribute to selection bias. It is possible that their slightly larger effect sizes were a result of the instrument used (Activity Inventory) or study design, or a combination of both. Discrepancies in effect sizes, of course, can also be indicative of noise in questionnaires: high effect sizes may mean that the questionnaire is exaggerating the effect after treatment, whereas low effect sizes may mean that the questionnaire is understating the real treatment effect. Such discrepancies make extrapolating increased scores in vision-related quality of life to patient's actual improvement difficult. Ideally, both the Impact of Vision Impairment and the National Eye Institute Visual Functioning Questionnaire need to be psychometrically evaluated for reproducibility and reliability in our clinic population. However, such analysis was beyond the scope of the study.

Predicting Change in Vision-related Quality-of-life Scores from Patient Characteristics and Occupational Therapy

Results from our multivariable logistic regression show that patient characteristics such as being male, visual field loss, and treatment options such as referral to occupational therapy and length of occupational therapy predict less increase in vision-related quality-of-life scores, whereas having a diagnosis other than macular disease predicts an increase in vision-related quality-of-life scores. Vision rehabilitation studies indicate that age rather than sex is a predictor of vision-related quality of life.[27,28] Most associations depend on sample size and distribution, and it is possible that sex was not a predictor in our study because of small sample size and the uneven distribution of our age group (our participants' median age was 75 years). Because sex is not an established predictor in vision rehabilitation, perhaps men's lower likelihood of increased scores on vision-related quality of life reflects the idiosyncrasy of our study population. Field loss, especially hemianopia and peripheral field loss, predicted a lower likelihood of positive change in scores on vision-related quality of life. Field loss is known to negatively affect vision-related quality of life in glaucoma and hemianopia patients.[29,30] In our study, higher scores on the Impact of Vision Impairment well-being subscale but not on the mobility subscale were reduced by visual field loss. However, this result should be interpreted with caution because there were only eight participants in the hemianopia and peripheral field loss group. Because the Impact of Vision Impairment was designed for patients with poor central vision, it is possible that it may not be sensitive enough to detect mobility-related vision-related quality-of-life change in patients with bad peripheral vision but good central vision.[31] Our findings about occupational therapy completion and duration are unique because, to our knowledge, no one has looked at occupational therapy as a predictor of vision-related quality of life. It certainly seems intuitive that those who are referred to occupational therapy but do not attend do worse on the Impact of Vision Impairment mobility subscale, as our occupational therapists do training about mobility issues such as fall prevention strategies. It is also reasonable that those who attend more than 3 hours of occupational therapy do not do well on the socioemotional National Eye Institute Visual Functioning Questionnaire subscale. Longer occupational therapy hours often indicate patients with greater need for rehabilitation or more rehabilitation goals. It is worth noting that the "not referred" occupational therapy group, which was taken as a reference group in the analyses, had only nine participants in it, and thus, the results should be interpreted with caution. The small sample size in the group likely led to large confidence intervals for the occupational therapy completion variable in Table 3. In our study, diagnoses other than macular disease predicted greater scores on the National Eye Institute Visual Functioning Questionnaire socioemotional subscale. Because it is likely that nonmacular diseases often encompass diseases of peripheral vision, it is counterintuitive that those with visual field loss have decreased scores on Impact of Vision Impairment well-being, an opposite result with the National Eye Institute Visual Functioning Questionnaire. This discrepancy is an example of a discordance between the Impact of Vision Impairment questionnaire and the National Eye Institute Visual Functioning Questionnaire.

Limitations

There are several limitations to our study results. First, our results may not be generalizable to other populations or other models of vision rehabilitation. There was variability in how soon patients completed questionnaires after rehabilitation, which may have confounded results; however, such inconsistency is inevitable in usual care, and no significant relationship between follow-up time and effect size was reported in a systematic review of outcomes of vision rehabilitation.[6] Second, those with better vision (i.e., better contrast sensitivity) were more likely to not respond to post-rehabilitation questionnaires. This may have created a sample bias in our study population. Those who consented for the study and those who did not consent, however, did not differ significantly, thus showing that our overall sample was representative of all patients attending the clinic. Third, an adaptive response questionnaire, one that estimates a patient's goals and his/her difficulty based on their relevance and importance to the patient, was not used in this study. The use of one such questionnaire, Activity Inventory, has been advocated in vision rehabilitation research.[25,27] These questionnaires are generally lengthy to administer and do not include a well-being subscale, which we find important as an outcome of vision rehabilitation; however, comparing the performance of a quality-of-life questionnaire and an Activity Inventory–type questionnaire may be of interest in future studies. Lastly, as mentioned previously, the questionnaires needed to be analyzed more thoroughly because discrepancies in effect sizes, as well as

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disagreement in some results, may imply questionnaire noise. Nonetheless, one strength of our study was our use of Rasch analysis to assess the psychometric properties of the Impact of Vision Impairment and the National Eye Institute Visual Functioning Questionnaire and to convert the raw, ordinal scores to interval level data. Rasch analysis improves measurement precision and ability to detect significant associations (Fenwick et al. 2013;[32] Garamendi et al. 2006;[33] Norquist et al. 2004[34]).

CONCLUSIONS

In this study, we found that vision-related quality-of-life scores increased significantly after our usual care comprehensive vision rehabilitation at Massachusetts Eye and Ear Infirmary using both the Impact of Vision Impairment and the National Eye Institute Visual Functioning Questionnaire as patient-reported outcomes. Treatment effect size was biggest using the Impact of Vision Impairment; however, it is not clear if that means that Impact of Vision Impairment is more sensitive to vision rehabilitation or tends to exaggerate the treatment effect. As hypothesized, certain clinically relevant patient characteristics and treatment options were predictors of vision-related quality of life. Male sex, having field loss, not completing occupational therapy, or having more hours of occupational therapy may all indicate patients in need of different rehabilitation intervention, as they had less likelihood of having increased vision-related quality-of-life scores. Most important findings for vision rehabilitation clinicians include poor follow-through with occupational therapy (57.8% did not complete or attend occupational therapy) and that many patient variables did not predict change in vision-related quality of life, perhaps implying that there are other variables we did not measure that may play a more significant role in vision-related quality of life, such as motivation. A study design that allows for comparison of treatment versus nontreatment groups such as a crossover design would be optimal to better assess the unbiased effect of usual care vision rehabilitation. Future studies should focus on psychometrically comparing the Impact of Vision Impairment and the National Eye Institute Visual Functioning Questionnaire after vision rehabilitation and including a control group to elucidate effectiveness of vision rehabilitation in a general population.[35]

ARTICLE INFORMATION

Submitted: February 7, 2018

Accepted: September 25, 2018

Funding/Support: None of the authors have reported funding/support.

Conflict of Interest Disclosure: None of the authors have reported a financial conflict of interest.

Author Contributions and Acknowledgments: Conceptualization: AS, MLJ; Data Curation: AS; Formal Analysis: EF, RM; Project Administration: AS; Supervision: MLJ; Writing – Review & Editing: WS, MLJ.

The authors would like to thank Ms. Rong Guo for her statistical assistance in the early preparation of the article.

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